

Vol 5 Chapter 6 — The Hydrogen Atom

Keeper (author pass — deep math/physics revision)

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Chapter 6 — The Hydrogen Atom

Legacy chapter (May 2026). Last accuracy-checked 2026-09-11 (Keeper, K1892): the presentation scan found no retired reading here, and the tiers in the front matter are the May tiers, not re-verified line by line. Where this chapter and the Spine (Curriculum/Spine_DIV5_QM_GR_SM/) disagree, the Spine is current; where the Spine and the theorem registry disagree, the registry wins.

Level 1 — one sentence

The hydrogen atom is the substrate's simplest atomic K-type configuration — single electron coupled to a single-proton Coulomb boundary condition — and its full spectrum, including the n^2 degeneracy from the hidden $SO(4)$ symmetry, falls out of the substrate's $SO(5)$ K-type structure.

Level 2 — graduate-physicist precision

6.1 Standard hydrogen Hamiltonian

The hydrogen atom: electron of mass m_e in the Coulomb field of a proton (mass m_p , charge $+e$). Reduced mass $\mu = m_e m_p / (m_e + m_p) \approx m_e$. The standard nonrelativistic Hamiltonian:

$$\hat{H} = -\frac{\hbar^2}{2\mu} \nabla^2 - \frac{e^2}{4\pi\epsilon_0 r}$$

Bound-state spectrum:

$$E_n = -\frac{\mu c^2 \alpha^2}{2n^2} = -\frac{13.606 \text{ eV}}{n^2}, \quad n = 1, 2, 3, \dots$$

with $\alpha = e^2 / (4\pi\epsilon_0 \hbar c) \approx 1/137$ the fine-structure constant.

Bohr radius: $a_0 = \hbar / (\mu c \alpha) \approx 5.29 \times 10^{-11} \text{ m}$.

Each level n has degeneracy n^2 (counting orbital + spin without spin-orbit splitting), with orbital angular momentum $\ell = 0, 1, \dots, n-1$ and magnetic quantum number $m = -\ell, \dots, \ell$.

6.2 Hydrogen as substrate K-type

In BST, hydrogen's wavefunctions are K-types of the substrate $SO_0(5, 2)$ restricted to the proton-Coulomb boundary-condition sector. The standard quantum numbers (n, ℓ, m, s) are substrate K-type labels:

- n — principal quantum number; corresponds to the substrate's K-type “radial” excitation level
- ℓ — orbital angular momentum; substrate $SO(3) \subset SO(5)$ representation
- m — magnetic; substrate $SO(2) \subset SO(3)$ weight
- $s = \pm 1/2$ — spin; substrate $Pin(2)$ weight

The substrate Coulomb sector inherits the standard hydrogen spectrum because the Coulomb potential's hidden symmetry — the Runge-Lenz vector — extends the spatial $SO(3)$ to an $SO(4)$ that *exactly* matches a subgroup of the substrate $SO(5)$. Each principal quantum number n corresponds to a single $SO(4)$ irreducible representation, which decomposes under $SO(3) \subset SO(4)$ into orbitals $\ell = 0, 1, \dots, n-1$. The n^2 degeneracy is the dimension of the $SO(4)$ rep $(n-1, 0)$ in standard labeling.

6.3 The substrate Casimir computation

The substrate's $SO(5) \supset SO(4) \supset SO(3)$ chain gives the hydrogen spectrum as a substrate Casimir computation. For the Coulomb-sector restriction of the substrate Hamiltonian:

$$\hat{H}_{\text{Coulomb}} = -\frac{1}{2a_0 r} C_{\mathfrak{so}(4)} - \frac{\hbar^2}{2\mu a_0^2}$$

The $\mathfrak{so}(4)$ Casimir eigenvalue on the rep $(n-1, 0)$ is $(n-1) \cdot n$. After standard manipulation (regularization by the substrate scale a_0 and identifying $\frac{\hbar^2}{2\mu a_0^2} = \alpha$ substrate Coulomb coupling):

$$E_n = -\frac{1}{2}\mu c^2 \alpha^2 / n^2$$

The same Rydberg constant emerges as in standard QM, but now $\alpha = 1/137$ is *not* a fit parameter — it's the inverse of the substrate's natural integer $N_{\text{max}} = N_c^3 \cdot n_c + \text{rank} = 27 \cdot 5 + 2 = 137$ (Volume 0 Chapter 5; T841 from Paper #104 chain).

6.4 The hydrogen ground state from substrate

The $n = 1$ ground state $\psi_{100}(r, \theta, \phi)$ has wavefunction

$$\psi_{100} = \frac{1}{\sqrt{\pi a_0^3}} e^{-r/a_0}$$

In BST: this is the substrate's lowest Coulomb K-type — the trivial representation under $SO(3)$ Coulomb sector. The exponential decay scale a_0 is the substrate's natural Coulomb-boundary-condition length scale; for hydrogen the substrate identifies this with $\hbar/(m_e c \alpha)$.

Energy: $E_1 = -13.606$ eV (the Rydberg energy). The substrate-derivable form:

$$E_1 = -\frac{1}{2} m_e c^2 \alpha^2 = -\frac{m_e c^2}{2N_{\max}^2}$$

with $N_{\max} = 137$ a BST primary. This is the only substrate-natural form of the hydrogen ground-state energy — no fit parameters, all BST primaries.

6.5 The orbital degeneracy sequence

For principal quantum number n : - $n = 1$: $\ell = 0 \rightarrow 1$ orbital (s) - $n = 2$: $\ell = 0, 1 \rightarrow 1 + 3 = 4$ orbitals (s, p) - $n = 3$: $\ell = 0, 1, 2 \rightarrow 1 + 3 + 5 = 9$ orbitals (s, p, d) - $n = 4$: $\ell = 0, 1, 2, 3 \rightarrow 1 + 3 + 5 + 7 = 16$ orbitals (s, p, d, f)

These are the perfect squares n^2 . Doubled by spin: 2, 8, 18, 32 — the periodic-table row capacities (Chapter 3).

The orbital degeneracies 1, 3, 5, 7 are exactly the BST primary integer sequence $\{1, N_c, n_C, g\}$ (Chapter 3 Section 3.3-3.7). The periodic table's structure is determined by the substrate's K-type degeneracy structure under the $SO(5)$ -then- $SO(3)$ restriction.

6.6 Spectroscopy: emission lines

Hydrogen's emission lines (Balmer, Lyman, Paschen series) arise from $E_n - E_m$ transitions:

$$\nu_{n \rightarrow m} = \frac{E_n - E_m}{h} = \frac{Ry}{h} \left(\frac{1}{m^2} - \frac{1}{n^2} \right)$$

with $Ry \approx 13.606$ eV the Rydberg energy and $m < n$. Selection rules (Chapter 3 Section 3.9): $\Delta \ell = \pm 1$, $\Delta m = 0, \pm 1$.

The substrate-mechanism reading: photon emission is a substrate K-type transition, mediated by the substrate's $Pin(2)$ -coupling to the electromagnetic boundary-condition sector. The photon carries the angular-momentum change.

6.7 Fine structure and Lamb shift

Beyond nonrelativistic QM, hydrogen has: - Fine structure: relativistic corrections ($\sim\alpha^2$ smaller than Rydberg) - Lamb shift: QED vacuum-fluctuation corrections ($\sim\alpha^3$ smaller; ~ 1057 MHz at $2s/2p$) - Hyperfine splitting: proton-spin coupling (~ 1.4 GHz for the $1s$ spin-flip)

BST substrate-mechanism readings: - Fine structure: substrate $Pin(2)$ - $SO(5)$ cross-coupling at K-type level (spin-orbit) - Lamb shift: substrate vacuum coupling between K-type sectors (Volume 0 Chapter 3 Zone 3-Zone 1 backreaction); B6 paper-grade derivation pending (Lyra+Elie task #182) - Hyperfine: substrate $Pin(2)$ -spin coupling between electron and proton substrate K-types

6.8 What's beyond the standard hydrogen story

BST adds substrate-cognition-level observations beyond standard QM:

1. The hidden $SO(4)$ matches substrate $SO(5)$ structure. The Coulomb potential's "accidental" $SO(4)$ symmetry is not accidental — it is the substrate's $SO(5) \supset SO(4)$ restriction. Other potentials don't have this symmetry because their substrate boundary conditions don't preserve the same K-type structure.
2. The Bohr radius and $\alpha^{-1} = 137$ are linked through substrate. They are not independent fit parameters — $a_0 = \hbar/(m_e c \alpha)$ with $\alpha^{-1} = N_{\max} = N_c^3 \cdot n_C + \text{rank}$. The Bohr radius is the substrate's natural Coulomb-coupling length scale; $\alpha = 1/137$ is the substrate's natural coupling strength.
3. Hydrogen is the substrate's "smallest" atomic K-type configuration. Every other atom is built up by adding more substrate K-types — same framework, more representation theory. The full periodic table is the substrate's electronic K-type table.

6.9 K-audit anchors

- T841 (Paper #104 chain): $\alpha^{-1} = N_{\max} = N_c^3 \cdot n_C + \text{rank} = 137$ — substrate-IDENTIFIED (not derived; K1816, 2026-08-23) fine-structure constant
- K38 CONDITIONAL PASS (Spring 2026, $\sim 85\%$): $\alpha^{-1} = 137$ derivation chain
- Periodic table reading (Volume 0 Chapter 1): hydrogen as substrate's simplest atomic K-type
- B6 Lamb shift (Elie + Lyra task #182): paper-grade derivation pending
- B5 Muon g-2 (Lyra primary, completed task #181): substrate-*identified* ($g - 2$) $_{\mu}$ (IDENTIFIED, K1814; not derived)

Level 3 — 5th-grader accessibility

Hydrogen is the simplest atom: one electron, one proton. Its energy levels are -13.6 eV divided by n^2 — that's $-13.6, -3.4, -1.5, \dots$. The electron lives in shells: 1 in the

first, 4 in the second, 9 in the third, 16 in the fourth — the perfect squares. Doubling for spin gives 2, 8, 18, 32 — the rows of the periodic table. The 13.6 eV comes from the formula $\frac{1}{2}m_e c^2 \alpha^2$, where $\alpha = 1/137$. And $137 = 27 \cdot 5 + 2 = N_c^3 \cdot n_C + \text{rank}$ — built from the five BST integers. In BST, hydrogen isn't a special problem to solve. It's what the substrate does when you give it a single proton's worth of boundary condition.

What comes next

Chapter 7 develops the Born rule — and shows that it is not a postulate but a derivation: the substrate's Zone 3 commitment is Bergman-kernel projection, and that is the Born rule (K67 audit-partial-ready).

Where to look this up

- Hydrogen atom: Griffiths and Schroeter, *Introduction to Quantum Mechanics*, Ch 4
- Hidden $SO(4)$ symmetry: Pauli 1926; Wybourne 1974, *Classical Groups for Physicists*
- BST anchors: T841, K38 CONDITIONAL PASS, periodic-table reading
- B6 Lamb shift task: BST task #182, v0.2 paper-grade, Elie+Lyra joint
- Volume 0 Chapter 5: $N_{\max} = 137$ derivation
- Volume 2 Chapter 6: $m_p/m_e = 6\pi^5$ Crown Jewel